Thomas Coen

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC**

B.S.E. Mechanical Engineering, Class of 2029 · Aerospace Engineering Certificate *GPA: 3.78*

Relevant coursework: Thermodynamics, Mechanics of Solids, Differential Equations, Engineering Design

TECHNICAL EXPERIENCE

Duke AERO (Rocketry Team) **Durham, NC**

Structures Lead *July 2026 – Present*

- Lead structural design for the team's 30,000 ft student researched and developed (SRAD) rocket for the 2027 International Rocket Engineering Competition (IREC) and a minimal two-stage staging tester
- Pushing the team's first year of filament winding: winding fiberglass and carbon flight tubes and tuning the machine, aiming for faster, cheaper, and potentially stronger tubes than hand layup; stress-strain characterization of samples next

Structures & Solid Propulsion Member *2025 – 2026*

- Machined 6061 aluminum nozzle casing on CNC lathe; hardware survived full-duration static hot fire and the launch at IREC 2026, where the rocket took 3rd in the 30,000 ft SRAD propulsion category
- Cut prepreg carbon fiber and performed layup for rocket body tubes
- Machined canard and air brake housing components on a Viwa CNC mill
- Assembled motor hardware components for static hot fire and flight
- Traveled to IREC 2026 in Midland, TX; supported competition operations and helped recover the vehicle after flight

BreakerBots, FIRST Robotics Competition Team 5104 **Pacific Grove, CA**

Co-Captain, CAD & Manufacturing Lead, Drive Coach *2022 – 2025*

- Redesigned Leviathan's 3-stage cascade elevator (9.5 ft travel, 2× Kraken X60, 1:5 reduction), the robot's main scoring cycle bottleneck: added constant force spring assist, cutting extension time from 3.5 s to 1.8 s for an estimated 25% more scoring cycles (2025)
- Cut Leviathan's weight from 128 lb to 114 lb by pocketing aluminum structure and replacing machined brackets with 3D printed carbon fiber, clearing the 115 lb competition limit (2025)
- Coordinated mechanical integration across the team's subteams as Co-Captain both seasons (2024–2025), including 5 subteams and 35 members on Leviathan
- Designed, CAD'd, and machined Typhoon's intake pivot and intake, balancing frame-perimeter packaging, pivot backlash, and mechanism weight; CAM'd and machined a large share of the robot's structural parts and grew the manufacturing subteam to 15+ machinists (2024)
- First year as Manufacturing Lead: grew the subteam from 2 to 8+ machinists and CAM'd/machined major structural components across Kraken and Cthulhu (2023)
- Machined structural components (CNC-milled aluminum tubing, bellypan, polycarbonate panels) as a first-year machinist on Charybdis (2022)
- Progressed from machinist (2022) to Manufacturing Lead & Drive Coach (2023) to Co-Captain & Manufacturing Lead (2024) to Co-Captain, CAD & Manufacturing Lead (2025)

Duke EGR 121 — Sunflower Seed Sheller **Durham, NC**

Mechanical Design *2026*

- Designed the rollers, drive mechanism, and funnel; did all CAD for a 4-person team
- Iterated roller geometry to crack each seed along its seam instead of crushing it at random, eliminating the kernel damage seen in early prototypes
- Produced 41 intact kernels in a 3-minute run — top score of 8 teams on a yield-minus-waste metric and 2nd among all first-year mechanical engineering teams

SKILLS

Design: SolidWorks, Onshape, Fusion 360 (CAM + CAD), AutoCAD

Manufacturing: CNC milling (Tormach 1100MX/770M, Viwa) and turning, manual Z-axis milling, manual lathe, 3D printing, carbon fiber and fiberglass layup, filament winding, laser cutting; Titans of CNC certification series